Problem

Title

Count Prime Numbers in Range [L, R]

Topics

Prime Numbers, Number Theory, Nested Loops, Conditional Statements, Range Iteration

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

During a competitive programming bootcamp, students are assigned a mathematical task related to number ranges. Instead of checking just one number, they now need to analyze an entire interval of numbers.

The trainer provides two integers **L** and **R**, representing the starting and ending values of a range (inclusive). The students must determine how many prime numbers exist within this range.

A prime number is defined as a number greater than 1 that has exactly two distinct divisors: 1 and itself.

The challenge requires students to:

- Iterate through every number between L and R.
- For each number, check whether it is prime.
- Use proper looping constructs (outer loop for range traversal).
- Use nested loop logic for divisor checking.
- Apply conditional structures correctly.
- Handle edge cases such as $L \leq 1$.

For example:

- In the range 1 to 10 → Prime numbers are 2, 3, 5, 7 → Total = 4
- In the range 10 to 20 → Prime numbers are 11, 13, 17, 19 → Total = 4

Students are not allowed to use built-in prime libraries. The logic must be implemented manually using loops and condition checking.

This problem strengthens:

- Nested loop thinking
- Range-based iteration
- Mathematical optimization
- Time complexity awareness

Your task is to write a program that reads L and R and prints the total count of prime numbers in the range [L, R].

Input Format

- Two space-separated integers:

```
L R
```

Constraints:

```
0 ≤ L ≤ R ≤ 10^6
```

Output Format

- Print a single integer representing the count of prime numbers between L and R (inclusive).

Examples

Example 1

Input:

```
1 10
```

Output:

```
4
```

Example 2

Input:

```
10 20
```

Output:

```
4
```

Example 3

Input:

```
20 30
```

Output:

```
2
```

(Prime numbers: 23, 29)

Example 4

Input:

```
0 5
```

Output:

```
3
```

(Prime numbers: 2, 3, 5)